

NexSMS —Step-by-Step Build Plan

Version: 1.0 **Status:** Final (awaiting build start) Order: 00000000 0000 0000 0000
0000 0000, 000000 0000 verify 000000

Phase 0 —Setup & Foundation

- ☐ 0.1 Create monorepo structure: frontend/ + backend/
- ☐ 0.2 Frontend: React + Vite + Tailwind CSS scaffold
- ☐ 0.3 Backend: Node.js + Express + PostgreSQL connection
- ☐ 0.4 Install PostgreSQL + Redis (local dev)
- ☐ 0.5 Database migrations setup (schema creation)
- ☐ 0.6 i18n skeleton (English + 中文 JSON files)
- ☐ 0.7 Theme system (CSS variables: colors, fonts) —admin-controllable

Exit criteria: npm run dev frontend+backend 00000, DB connected0

Phase 1 —Auth & User Management

- ☐ 1.1 User registration + login (JWT)
- ☐ 1.2 Admin registration + login
- ☐ 1.3 Role-based access (user / admin / super admin)
- ☐ 1.4 Password hashing (bcrypt) + reset
- ☐ 1.5 2FA OTP for admin login
- ☐ 1.6 Session/refresh token
- ☐ 1.7 i18n: auth pages English + 中文

Exit criteria: user 0 admin 000000 login, protected routes work 00000

Phase 2 —Chat Dashboard (Google Voice UI)

- ☐ 2.1 Layout: left inbox/thread list + right conversation pane
- ☐ 2.2 Thread list: contact name/number, preview, timestamp, unread badge
- ☐ 2.3 “+ New” button → new conversation (enter number)
- ☐ 2.4 Conversation view: chat bubbles (sent/received), date separators
- ☐ 2.5 Message composer: input + send button, “From number” selector
- ☐ 2.6 Auto-scroll, optimistic UI, typing indicator (optional)
- ☐ 2.7 Responsive: mobile collapses to single pane
- ☐ 2.8 i18n: chat UI English + 中文

Exit criteria: UI 000000000 Google Voice-like, mock data 000000 0000 00000

Phase 3 —Provider Abstraction Layer (Multi-Provider)

- ☐ 3.1 Common interface: sendSms(), receiveWebhook(), checkNumber(), buyNumber()
- ☐ 3.2 Twilio adapter (0000000)
- ☐ 3.3 SingleHouse adapter
- ☐ 3.4 Plivo adapter
- ☐ 3.5 Telnyx adapter
- ☐ 3.6 Provider registry + routing by priority
- ☐ 3.7 Provider health check

- ☐ 3.8 Failover: primary down → next provider (auto/manual)
- ☐ 3.9 Webhook receiver per provider (signature validation)

Exit criteria: 1. send API, 2. provider 3. 4. 5. 6. 7. 8. 9. 10. 11. 12. 13. 14. 15. 16. 17. 18. 19. 20. 21. 22. 23. 24. 25. 26. 27. 28. 29. 30. 31. 32. 33. 34. 35. 36. 37. 38. 39. 40. 41. 42. 43. 44. 45. 46. 47. 48. 49. 50. 51. 52. 53. 54. 55. 56. 57. 58. 59. 60. 61. 62. 63. 64. 65. 66. 67. 68. 69. 70. 71. 72. 73. 74. 75. 76. 77. 78. 79. 80. 81. 82. 83. 84. 85. 86. 87. 88. 89. 90. 91. 92. 93. 94. 95. 96. 97. 98. 99. 100. 101. 102. 103. 104. 105. 106. 107. 108. 109. 110. 111. 112. 113. 114. 115. 116. 117. 118. 119. 120. 121. 122. 123. 124. 125. 126. 127. 128. 129. 130. 131. 132. 133. 134. 135. 136. 137. 138. 139. 140. 141. 142. 143. 144. 145. 146. 147. 148. 149. 150. 151. 152. 153. 154. 155. 156. 157. 158. 159. 160. 161. 162. 163. 164. 165. 166. 167. 168. 169. 170. 171. 172. 173. 174. 175. 176. 177. 178. 179. 180. 181. 182. 183. 184. 185. 186. 187. 188. 189. 190. 191. 192. 193. 194. 195. 196. 197. 198. 199. 200. 201. 202. 203. 204. 205. 206. 207. 208. 209. 210. 211. 212. 213. 214. 215. 216. 217. 218. 219. 220. 221. 222. 223. 224. 225. 226. 227. 228. 229. 230. 231. 232. 233. 234. 235. 236. 237. 238. 239. 240. 241. 242. 243. 244. 245. 246. 247. 248. 249. 250. 251. 252. 253. 254. 255. 256. 257. 258. 259. 260. 261. 262. 263. 264. 265. 266. 267. 268. 269. 270. 271. 272. 273. 274. 275. 276. 277. 278. 279. 280. 281. 282. 283. 284. 285. 286. 287. 288. 289. 290. 291. 292. 293. 294. 295. 296. 297. 298. 299. 300. 301. 302. 303. 304. 305. 306. 307. 308. 309. 310. 311. 312. 313. 314. 315. 316. 317. 318. 319. 320. 321. 322. 323. 324. 325. 326. 327. 328. 329. 330. 331. 332. 333. 334. 335. 336. 337. 338. 339. 340. 341. 342. 343. 344. 345. 346. 347. 348. 349. 350. 351. 352. 353. 354. 355. 356. 357. 358. 359. 360. 361. 362. 363. 364. 365. 366. 367. 368. 369. 370. 371. 372. 373. 374. 375. 376. 377. 378. 379. 380. 381. 382. 383. 384. 385. 386. 387. 388. 389. 390. 391. 392. 393. 394. 395. 396. 397. 398. 399. 400. 401. 402. 403. 404. 405. 406. 407. 408. 409. 410. 411. 412. 413. 414. 415. 416. 417. 418. 419. 420. 421. 422. 423. 424. 425. 426. 427. 428. 429. 430. 431. 432. 433. 434. 435. 436. 437. 438. 439. 440. 441. 442. 443. 444. 445. 446. 447. 448. 449. 450. 451. 452. 453. 454. 455. 456. 457. 458. 459. 460. 461. 462. 463. 464. 465. 466. 467. 468. 469. 470. 471. 472. 473. 474. 475. 476. 477. 478. 479. 480. 481. 482. 483. 484. 485. 486. 487. 488. 489. 490. 491. 492. 493. 494. 495. 496. 497. 498. 499. 500. 501. 502. 503. 504. 505. 506. 507. 508. 509. 510. 511. 512. 513. 514. 515. 516. 517. 518. 519. 520. 521. 522. 523. 524. 525. 526. 527. 528. 529. 530. 531. 532. 533. 534. 535. 536. 537. 538. 539. 540. 541. 542. 543. 544. 545. 546. 547. 548. 549. 550. 551. 552. 553. 554. 555. 556. 557. 558. 559. 560. 561. 562. 563. 564. 565. 566. 567. 568. 569. 570. 571. 572. 573. 574. 575. 576. 577. 578. 579. 580. 581. 582. 583. 584. 585. 586. 587. 588. 589. 590. 591. 592. 593. 594. 595. 596. 597. 598. 599. 600. 601. 602. 603. 604. 605. 606. 607. 608. 609. 610. 611. 612. 613. 614. 615. 616. 617. 618. 619. 620. 621. 622. 623. 624. 625. 626. 627. 628. 629. 630. 631. 632. 633. 634. 635. 636. 637. 638. 639. 640. 641. 642. 643. 644. 645. 646. 647. 648. 649. 650. 651. 652. 653. 654. 655. 656. 657. 658. 659. 660. 661. 662. 663. 664. 665. 666. 667. 668. 669. 670. 671. 672. 673. 674. 675. 676. 677. 678. 679. 680. 681. 682. 683. 684. 685. 686. 687. 688. 689. 690. 691. 692. 693. 694. 695. 696. 697. 698. 699. 700. 701. 702. 703. 704. 705. 706. 707. 708. 709. 710. 711. 712. 713. 714. 715. 716. 717. 718. 719. 720. 721. 722. 723. 724. 725. 726. 727. 728. 729. 730. 731. 732. 733. 734. 735. 736. 737. 738. 739. 740. 741. 742. 743. 744. 745. 746. 747. 748. 749. 750. 751. 752. 753. 754. 755. 756. 757. 758. 759. 760. 761. 762. 763. 764. 765. 766. 767. 768. 769. 770. 771. 772. 773. 774. 775. 776. 777. 778. 779. 780. 781. 782. 783. 784. 785. 786. 787. 788. 789. 790. 791. 792. 793. 794. 795. 796. 797. 798. 799. 800. 801. 802. 803. 804. 805. 806. 807. 808. 809. 810. 811. 812. 813. 814. 815. 816. 817. 818. 819. 820. 821. 822. 823. 824. 825. 826. 827. 828. 829. 830. 831. 832. 833. 834. 835. 836. 837. 838.

Phase 4 – Number Pool & Assignment

- ☐ 4.1 numbers table: provider, geo/area code, status
- ☐ 4.2 Buy numbers (manual entry / provider API fetch)
- ☐ 4.3 Pool view: available / assigned / blocked
- ☐ 4.4 Assign number to user (admin manual + user self-service optional)
- ☐ 4.5 Multiple numbers per user + primary flag
- ☐ 4.6 Revoke / swap / pause numbers
- ☐ 4.7 Geo-picker (filter by area code/country)

Exit criteria: admin number [] [] [] [] pool-[] [] [] [] , user-[] [] assign [] [] , user-[] [] dashboard-[] num-
ber [] [] [] [] [] [] [] []

Phase 5 –Messaging Engine (Send/Receive)

- ☐ 5.1 Send SMS through provider layer (assigned number)
- ☐ 5.2 Inbound webhook → create/continue conversation thread
- ☐ 5.3 Per-conversation thread management (from number + contact)
- ☐ 5.4 Delivery status tracking: sent → delivered → failed
- ☐ 5.5 Auto-retry on failure
- ☐ 5.6 Per-number daily limit counter (24h window, DB)
- ☐ 5.7 Block message when daily limit reached (“Try again in X hours”)
- ☐ 5.8 SMS cost calculation + wallet deduction
- ☐ 5.9 Rate limiting per user (spam control)
- ☐ 5.10 A2P 10DLC compliance status field per number

Exit criteria: real SMS $\leq 1000000/1000000$, thread- ≤ 10000 , cost ≤ 10000 , daily limit ≤ 10000

Phase 6 —Billing System (2 Modes)

- ☐ 6.1 Plans table: Starter / Standard / Premier (quota + price)
- ☐ 6.2 Prepaid credits mode: wallet balance → per-SMS deduction
- ☐ 6.3 Subscription mode: monthly plan + SMS quota
- ☐ 6.4 Hybrid: quota exhausted → wallet charge OR block (admin rule)
- ☐ 6.5 Admin global billing-mode toggle
- ☐ 6.6 Per-user billing mode override
- ☐ 6.7 Per-plan daily SMS limits (per number)
- ☐ 6.8 Pay-per-SMS separate daily limit (admin-set)
- ☐ 6.9 Low-balance alert (email/in-app)

Exit criteria: admin mode 000000 user-00 0000 000 option-0 0000 0000, 000000 00000000 000000

Phase 7 —Admin Panel

- ☐ 7.1 Admin dashboard layout (sidebar + main)
- ☐ 7.2 Providers page: CRUD, credentials, enable/disable, priority, health
- ☐ 7.3 Users page: list, search, block/unblock, edit, password reset
- ☐ 7.4 Numbers page: pool, assign/revoke/swap, geo-picker, daily usage
- ☐ 7.5 Wallet page: credit/debit, transaction log, payment history
- ☐ 7.6 Payments page: gateway CRUD, enable/disable, fees, priority
- ☐ 7.7 Billing page: mode toggle, plan quotas, daily limits
- ☐ 7.8 Theme page: color, font, logo, branding (DB-driven)
- ☐ 7.9 Webhooks page: manager + signature validation
- ☐ 7.10 Audit log page
- ☐ 7.11 Analytics page: platform usage, provider cost report
- ☐ 7.12 Export/backup (CSV/JSON)
- ☐ 7.13 IP whitelisting for admin

Exit criteria: admin panel ☐☐☐☐ ☐☐☐☐☐ ☐ ☐☐☐ ☐☐☐☐

Phase 8 —Payment Gateways

- ☐ 8.1 Payment orders table + order flow
- ☐ 8.2 Binance Pay integration
- ☐ 8.3 Bybit integration
- ☐ 8.4 Stripe integration
- ☐ 8.5 Cryptomus integration
- ☐ 8.6 TRC20 (USDT) —wallet address payment
- ☐ 8.7 BEP20 (USDT) —wallet address payment
- ☐ 8.8 BTC —wallet address payment
- ☐ 8.9 BenlyPay integration
- ☐ 8.10 Gateway enable/disable respected in user payment page
- ☐ 8.11 Payment success → wallet credit (per gateway rules)
- ☐ 8.12 QR code: auto-generate from wallet address (TRC20/BEP20/BTC) + scan-to-pay
- ☐ 8.13 QR mode per gateway: Auto (API) / Manual upload / Off (admin-set)

Exit criteria: user ☐☐☐☐☐ enabled gateway ☐☐☐☐ ☐☐☐☐☐ → wallet credit ☐

Phase 9 —Advanced User Features

- ☐ 9.1 SMS scheduling (send later)
- ☐ 9.2 Bulk blast tool (with compliance/opt-in checks)
- ☐ 9.3 AI auto-reply (incoming → AI reply)
- ☐ 9.4 Smart inbox (spam/sales filter)
- ☐ 9.5 Reply suggestions
- ☐ 9.6 User API access + API key management
- ☐ 9.7 User webhooks (delivery callbacks)
- ☐ 9.8 Analytics dashboard (user-side: volume, cost, delivery rate)
- ☐ 9.9 Message template library
- ☐ 9.10 MMS/attachment support (premium)
- ☐ 9.11 Referral / discount system (invite → credit bonus)
- ☐ 9.12 Multi-currency wallet (USD + USDT + regional, admin exchange rates)
- ☐ 9.13 Public rate card page
- ☐ 9.14 KYC verification tier (unverified/verified limits)

Exit criteria: ☐☐ advanced feature user-☐☐ dashboard-☐ ☐☐☐ ☐☐☐☐

Phase 10 —Fraud Prevention & Feature Toggles

- ☐ 10.1 Flash USDT / fake crypto detection (contract check + explorer verify + min confirmations)
- ☐ 10.2 Carding prevention (3D Secure, AVS/CVV, velocity limit, VPN detection, chargeback monitor)
- ☐ 10.3 Risk scoring per payment + payment hold policy
- ☐ 10.4 Admin review queue for flagged payments
- ☐ 10.5 Replay protection + order expiry (24h auto-cancel)
- ☐ 10.6 Webhook IP whitelist + signature verification
- ☐ 10.7 Global IP blocklist
- ☐ 10.8 Message content filter (admin keyword list)
- ☐ 10.9 Number expiry & renewal (grace period)
- ☐ 10.10 Scheduled maintenance mode
- ☐ 10.11 Admin alert center (provider down, big payment, low balance, suspicious)
- ☐ 10.12 Performance optimizations (bulk queue, indexes, pagination)
- ☐ 10.13 **Feature Toggles page:** ☐ feature-☐ ON/OFF switch (admin-☐ ☐ ☐ ☐ control)

Exit criteria: ☐ ☐-☐ module ☐ ☐, ☐ ☐ admin toggle ☐ on/off ☐ ☐

Phase 11 —Homepage & Polish

- ☐ 11.1 Homepage: hero + trust bar + features + pricing + FAQ + CTA + footer
- ☐ 11.2 Homepage animations (fade-up, scroll reveal, glass navbar)
- ☐ 11.3 Theme applied site-wide from admin
- ☐ 11.4 Responsive design (mobile/tablet/desktop)
- ☐ 11.5 i18n complete: English + 中文 (all pages)
- ☐ 11.6 Loading states, empty states, error states (UX polish)
- ☐ 11.7 SEO meta tags + favicon + logo
- ☐ 11.8 Final QA + bug fixes

Exit criteria: ☐ ☐ platform production-ready look + everything working☐

Milestone Summary

Milestone	☐	Result
M1	Phase 0-2	UI skeleton + chat dashboard (mock)
M2	Phase 3-5	Real SMS + multi-provider + billing core
M3	Phase 6-8	Billing modes + admin panel + payments + QR
M4	Phase 9-10	Advanced features + fraud prevention + toggles
M5	Phase 11	Homepage + polish + i18n complete